Conjecture 00000001004: a q^4-point cap in PG$(6, q)$

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1 The fixed-dimension contradiction

The conjecture specifies $c_6 = 2$ and $\lceil 6/2 \rceil = 3$, so its $n = 6$ case asserts

$$m_2(6, q) = 2q^3\big(1 + O(q^{-1})\big).$$

We construct a genuine projective cap of size q^4 over every finite field K of odd characteristic, where $q = |K|$. Hence

$$m_2(6, q) \geq q^4.$$

There are arbitrarily large odd prime fields. The lower bound rules out even an additive $O(q^2)$ error around $2q^3$, and therefore the specified relative $O(q^{-1})$ expansion. A cap means that no three pairwise distinct points are collinear; no completeness or maximality condition is imposed on the constructed cap. The separate maximum m_2 is taken over all caps.

2 The actual projective construction

Choose a nonsquare $d \in K$. Let $Q(x, y) = x^2 - dy^2$. If $Q(x, y) = 0$ and $y \neq 0$, then $d = (x/y)^2$, a contradiction. If $y = 0$, then $x^2 = 0$ forces $x = 0$. Thus Q is anisotropic. For $u = (u_0, u_1, u_2, u_3) \in K^4$, put

$$v(u) = \big(1, u_0, u_1, Q(u_0, u_1), u_2, u_3, Q(u_2, u_3)\big) \in K^7$$

and set $S = \{[v(u)] : u \in K^4\} \subset \mathrm{PG}(6, K)$. The first coordinate is one, so $v(u) \neq 0$ and $[v(u)] = [v(v)]$ forces the projective scaling factor to be one. The remaining coordinates then give $u = v$. Consequently $|S| = q^4$. This is the Cartesian product of two anisotropic quadratic graph caps in affine three-space, embedded in the affine chart of projective six-space.

Suppose three distinct points of S are collinear. The two distinct first points span their projective line, so their representing vectors satisfy

$$v(w) = a\, v(u) + b\, v(v), \qquad a + b = 1.$$

If $a = 0$ or $b = 0$, the third point equals one of the first two. Otherwise, the elementary identity

$$Q(ax + bs, ay + bt) = aQ(x, y) + bQ(s, t) - abQ(x - s, y - t)$$

for $a + b = 1$, applied to both quadratic coordinate blocks, gives

$$abQ(u_0 - v_0, u_1 - v_1) = 0, \quad abQ(u_2 - v_2, u_3 - v_3) = 0.$$

Anisotropy yields $u = v$, again a contradiction. Thus S is a cap.

3 The true maximum and the asymptotic quantifiers

For the actual finite set of points of $\mathrm{PG}(6, K)$, take all subsets satisfying the cap condition. This family is nonempty because it contains the empty cap. A member of maximum cardinality exists; Lean defines `maxCapSet` by choosing such a member, and defines `maxSize` as its cardinality. It proves both that this chosen set is a cap and that every cap has at most `maxSize` points. The value is not assigned the constructed size. Applying its universal maximality property to S proves $q^4 \leq m_2(6, q)$.

The formal asymptotic domain `OddPrime` consists of actual primes $p \geq 3$, ordered by their natural values. `primeMax` is the maximum above for the actual prime field `ZMod p`. Euclid's theorem supplies arbitrarily large members. Lean proves

$$\forall C \in \mathbf{R}\ \forall N \in \mathbf{N}\ \exists p \geq N, \quad p \text{ an odd prime}, \quad 2p^3 + Cp^2 < m_2(6, p).$$

Indeed, choose p sufficiently larger than $C + 4$. Then $p^2 > 2p + C$ and multiplication by $p^2 > 0$ gives $p^4 > 2p^3 + Cp^2$. The constructed cap supplies the remaining inequality. This includes arbitrary real error constants, not just one numerical choice or a finite list of field sizes.

Using Mathlib's actual `IsBigO` and `atTop` filter on `OddPrime`, `no_quadratic_error` proves that $m_2(6, p) - 2p^3$ is not $O(p^2)$. The final theorem `no_claimed_expansion` directly rules out existence of a real function $r(p) = O(p^{-1})$ for which $m_2(6, p) = 2p^3(1 + r(p))$ eventually. The conjectured expansion through all prime powers would in particular hold on the unbounded odd-prime subfamily, so this disproves its fixed $n = 6$ necessary case.

4 Standard construction and formal verification

The product construction is classical. For definitions, affine-to-projective embedding and this family of lower bounds, see Elsholtz and Lipnik, *Exponentially Larger Affine and Projective Caps*, Introduction and Related Work. That discussion also gives stronger known bounds in dimension six. We use only the simpler q^4 lower bound and prove it in Lean, without assuming a published cap-size theorem.

Formal points are Mathlib's `Projectivization K (Fin 7 → K)`. `IsCap` literally excludes three distinct points satisfying Mathlib's `Projectivization.IsCollinear`. Its actual subspace and dimension definition is connected to the explicit vector-combination argument by the proved uniqueness of the line through two distinct points. Nonsquares are supplied by a proved finite-field theorem.

Lean 4.33.0 and Mathlib commit `db584cd6d46c92f209a44c0f1c829460d327499d` are pinned. Source-bound receipts record a new Main build, seventeen transitive axiom audits and full official `leanchecker --fresh Main` replay. No `sorry`, `native_decide` or added axiom is used. OpenAI Codex assisted, and a separate supervising Codex agent reviewed the formal chain and report. This is AI review, not human certification. Organizer acceptance is pending.